Week 6: Daily Journals

Monday:

We initiated integration testing by validating the X-ray sensor payload. We first established a baseline by testing the sensor in a direct configuration without the switchboard, which successfully yielded accurate data. Next, we integrated the switchboard into the data path; however, the initial telemetry was corrupted. A systematic inspection of the signal path revealed a cold/loose solder joint. After resoldering the joint, we re-verified the baseline configuration and subsequently confirmed that the X-ray sensor functioned perfectly through the switchboard, validating payload readiness.

Tuesday:

Today's objective was to validate the Software Defined Radio (SDR) subsystem. We fabricated a dedicated power distribution cable interfacing the power supply, Raspberry Pi 4, and the SDR to test the payload independently of the switchboard. Initial execution successfully generated a valid .iq file, establishing our hardware baseline. We then routed the SDR through the switchboard. During testing, we identified a power routing anomaly where current was drawing directly from the Pi rather than passing through the switchboard bus. Once this power isolation issue was resolved, the SDR successfully maintained functionality, fully verifying the subsystem.

Wednesday:

With the SDR and X-ray payloads fully verified, I shifted focus to the imaging subsystem, specifically testing the Arducam Starvis IMX585 (630 nm camera) connected directly to Raspberry Pi 4. After achieving consistent standalone performance, we modified an existing Y-cable to separate the power and data lines to mitigate the previous week's current limitations. We then introduced a USB 3.0 splitter to accommodate a dual-camera configuration. However, the system failed to deliver concurrent power to both sensors, and all image capture attempts under this architecture were unsuccessful.

Thursday:

I continued troubleshooting the imaging payload by reverting to a simplified hardware topology. I bypassed the USB 3.0 splitter and connected a single camera directly to the custom Y-cable; however, initialization still failed. I paused technical troubleshooting to attend a final team alignment meeting to establish the parameters for our capstone evaluation. The final test protocol requires the integrated system to sequentially collect data from each individual instrument one at a time. Following the meeting, we resumed camera debugging but were unable to resolve the power drop before concluding operations for the day.

Friday:

Today marked the conclusion of the 2026 Summer Research Internship. Before initiating our final capstone test, we conducted a final review of the 630 nm camera anomalies. While a direct Pi-to-camera connection functions reliably, introducing inline splitters or extended cabling causes critical voltage drops. We concluded that the cumulative insertion loss and power draw exceeded our baseline thresholds; therefore, we optimized the final FlatSat configuration to utilize a direct connection for the 630 nm camera. After fine-tuning a minor calibration issue on the X-ray sensor, we executed our final end-to-end integration test. The sequence was a complete success, with all flight software executing nominally, officially concluding the research program.

Pictures

Figure 1: FlatSat Testing

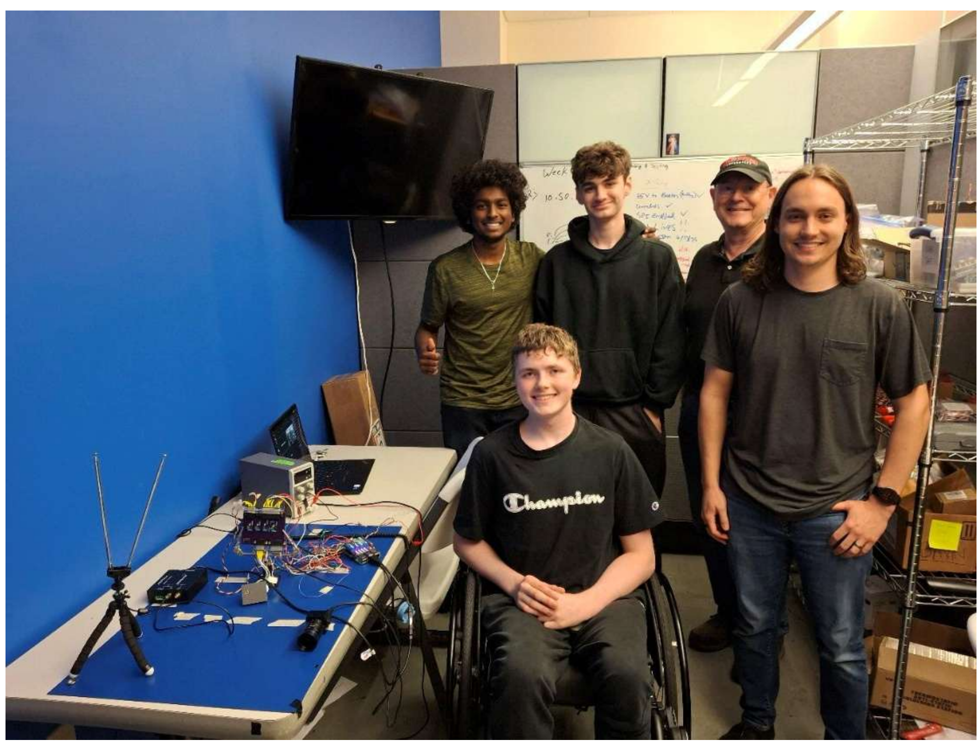

Figure 2: Summer Research Group